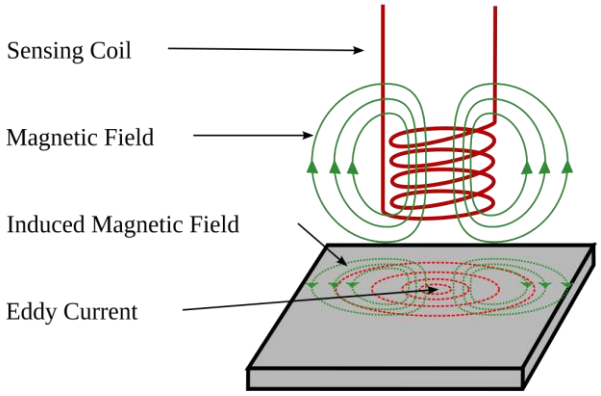
**Chapter 51 - Applications for Electromagnetic Induction**
**Calculus Physics II**

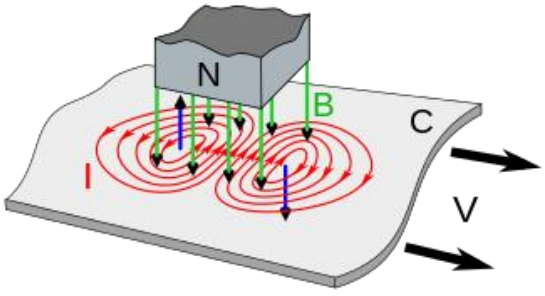
**Eddy Currents** - are electrical currents set up in bulk conductors experiencing a changing external magnetic field. Tthese induced currents produce a magnetic force that opposes the motion of the conductor because the currents are in a magnetic field. This induced force takes energy away from the object that is producing them. The Eddy Currents produce forces that oppose the motion of the object.

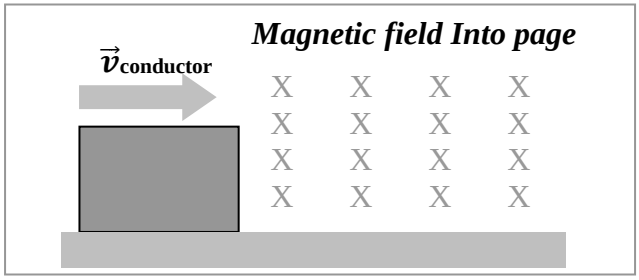
**Source of Eddy Currents** – Induced currents appear in solid conductors when the conductor is exposed to a changing magnetic field (changing magnetic flux). If the electrical resistance of a solid piece of conductor is low, the induced currents will be very large, this will result in substantial energy dissipation.

- The induced current produces a magnetic field that opposes the changing external field; therefore, it produces a field in the opposite direction as the external field. The presence of these Eddy currents makes it difficult to move a conductor through a magnetic field.
- The energy that creates the currents is taken from the kinetic energy of the moving conductor. Therefore, the conductor will slow down if no other energy is added to the conductor.



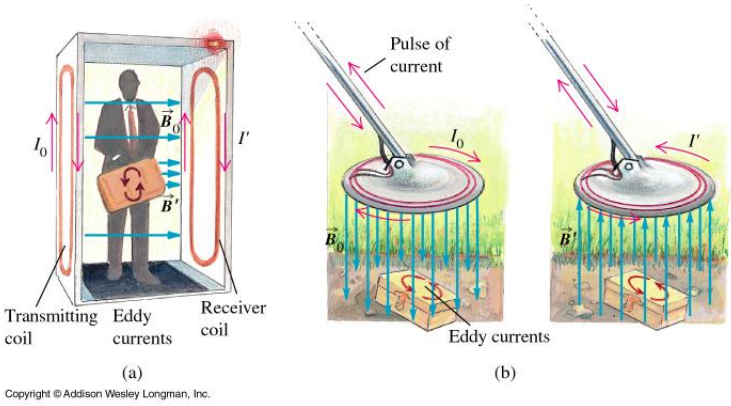

Example: The diagram shows a conductor sliding along a frictionless table when it encounters a uniform magnetic field. As the conductor enters the magnetic field, Eddy currents are induced in the conductor by the changing magnetic field, these induced currents create a magnetic field that opposes the change in the external field. The energy that creates the induced current comes from the kinetic energy of the conductor as it slides. The faster the conductor is moving the larger the Eddy currents and the more kinetic energy is lost from the conductor until it will eventually stop. Once the conductor is stopped there is no more motion to create the Eddy currents, so they disappear.



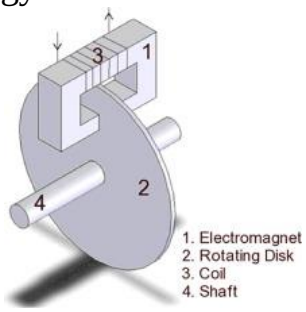
# Some Applications of Eddy Currents

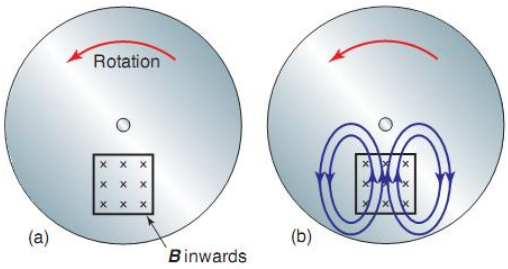
**Metal detectors** use a coil of wire with a current running through it, when a metal object is near, eddy currents are induced in the metal object along with an associated magnetic field from the conduction object, and this induced magnetic field is sensed by a second coil in the detector.



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**Magnetic brakes** When the electromagnet is engaged it produces a magnetic field that induces eddy currents in the rotating conducting disk. These eddy currents produce a magnetic field that opposes the field from the electromagnet. The magnetic force produced from the eddy currents magnetic field slows the rotation of the disk. This transfer mechanical energy (motion) to magnetic energy.




**Motion Dampening** – By exposing a conductor in motion to a fixed magnetic field, the motion of that conductor is reduced due to the eddy currents creating a force that counters its motion, this induced eddy current will slow the conductor to a stop over time.

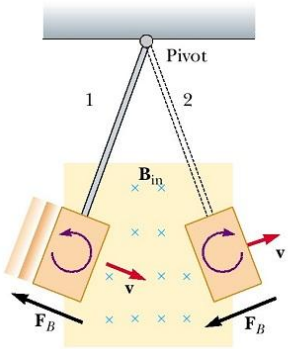
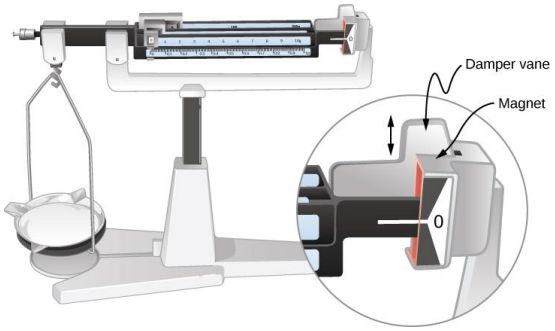
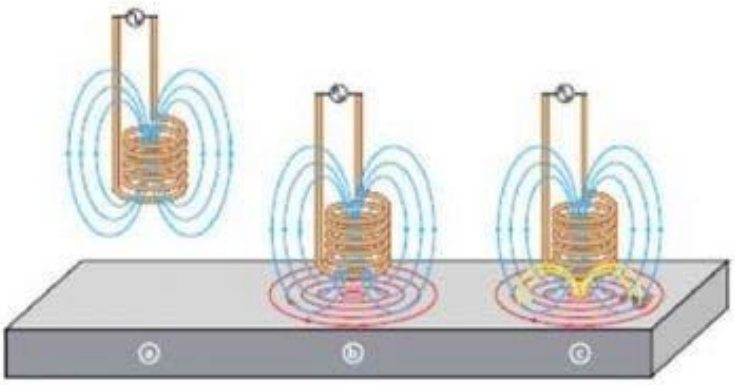



***Figure 31.23*** As the conducting plate enters the field (position 1), the eddy currents are counterclockwise. As the plate leaves the field (position 2), the currents are clockwise. In either case, the force on the plate is opposite the velocity, and eventually the plate comes to rest.



**Metal/conductor fault sensor** – Eddy currents are used to sense fauts and inhomogenaties in conducting material by sensing the induded magnetic field and if there is achange in that field. When the induced field changes, this indicates a fault.

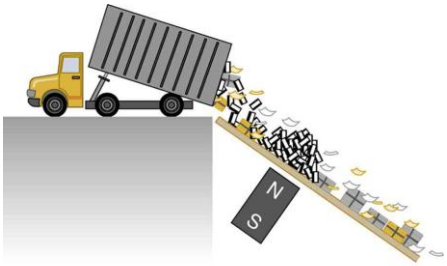


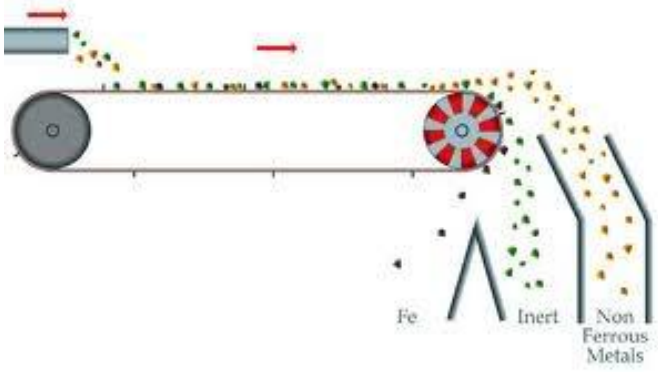
a—The alternating current flowing through the coil at a chosen frequency generates a magnetic field around the coil.

b—When the coil is placed close to an electrically conductive material, eddy current is induced in the material.

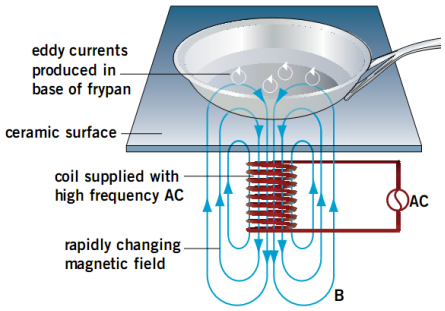
c—If a flaw in the conductive material disturbs the eddy current circulation, the magnetic coupling with the probe is changed and a defect signal can be read by measuring the coil impedance variation.

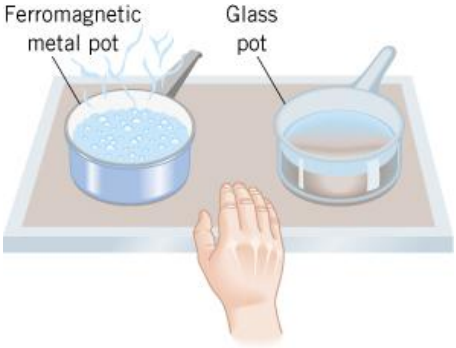
**Metal/Non-metal separator** – A magnetic field is used on a conveyer belt to induce eddy currents in metals that slow down their movement, but the magnetic field has no effect on non-metals therefore the conducting material is forced into a different path than the non-conducting materials.



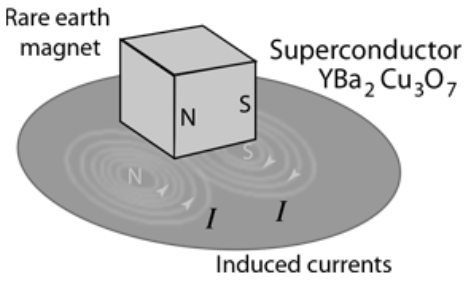



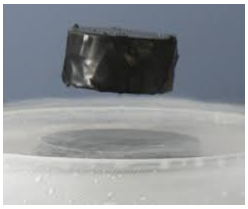

**Induction Stove** uses a changing magnetic field produced by a wire coil to induce eddy currents in metal pots. The induced currents experience a resistance that produces the heat on the bottom of the pot for cooking.



**Super Conductor levitation** – a magnet is levitated above a superconductor by eddy currents induced by the magnet in the superconducting material, once these currents are set up, there is no resistance to the current flow, they keep flowing and produce a magentic field that oppoeses the external field that created it, the magnet "floats" above the super conductor.

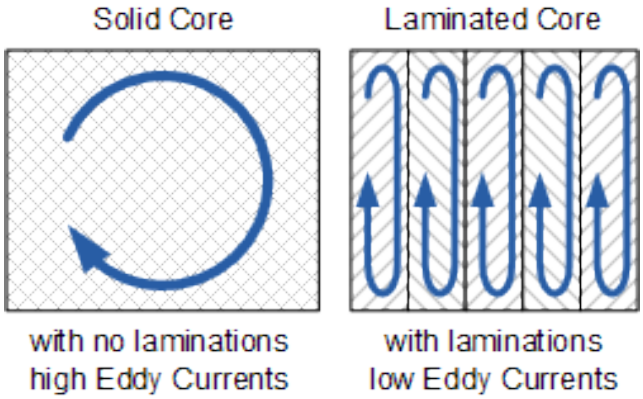




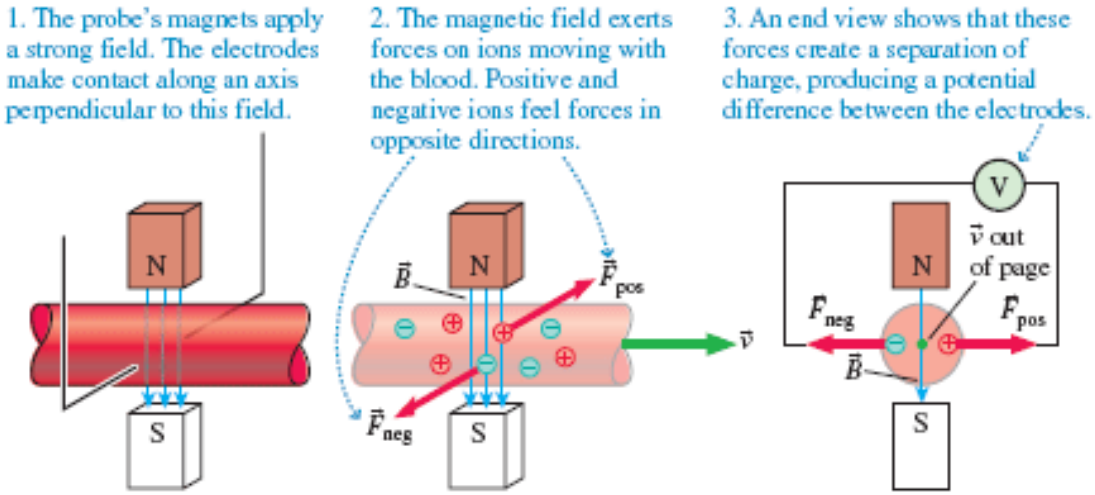

**Problems caused by Eddy currents**: Loss of energy in solid conductors in electric motor armatures due to motion in magnetic field. Loss of energy in large diameter wires.
Eddy currents cause unwanted problems by converting electrical or kinetic energy into heat, leading to power losses and reduced efficiency in devices like transformers and motors. They can also induce image artifacts and tissue heating in Magnetic Resonance Imaging (MRI) systems and cause physical braking or opposition to motion. Techniques to mitigate these problems include using laminated cores and ferrite materials to disrupt current loops

- Reduced efficiency: Eddy currents circulate within the conductive cores of transformers, motors, and inductors, dissipating electrical energy as heat. This results in lower efficiency, as more input power is needed to drive both the load and the eddy currents.
- Energy loss: The heat generated by eddy currents represents a loss of useful energy, increasing operating costs and potentially requiring larger components to handle the extra power input.

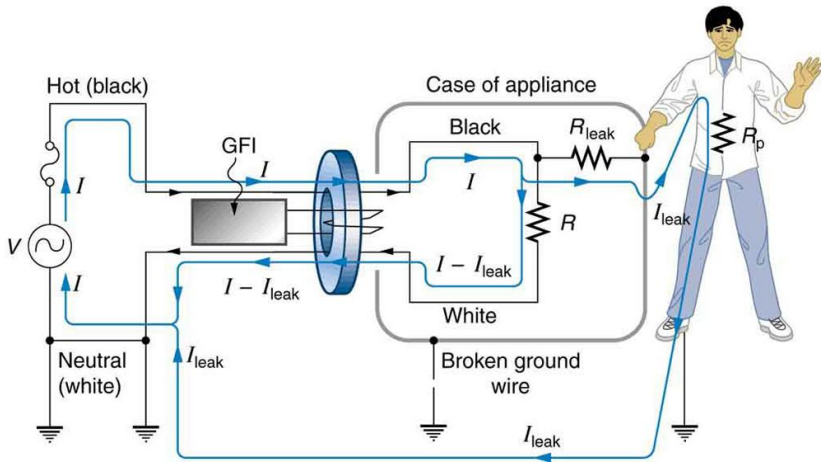


# Other Applications for the Electric Force and Electromagnetic Induction

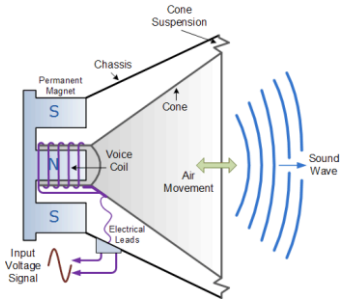
**Magnetic Flow meter** - Most fluids are electrical conductors (water, blood, oil, etc. …), when these fluids are moving through a magnetic field the charge in the fluids experience a force, this magnetic force pushes the positive change in one direction and the negative charge in the opposite, this is motional electromagnetic induction. The charge separation creates a voltage that can be measured, this induced voltage is proportional to the velocity of the fluid.




**GFI (Ground Fault Interrupter) Plugs for safety** – The "hot" and "neutral" wire of an electrical cord are run through a coil of wire (inductor), the current through the wire procures a changing magnetic field, because the return wire is there, the fields cancel and there is no induced voltage in the coil. If external to the plug the wire is grounded, there will be no return current through the plug only the outgoing current, this will induce a voltage in the coil and the circuit will be turned off.

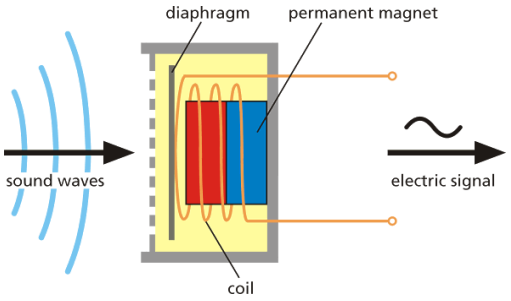


**Speakers and Microphone** - Both speakers and microphones work on the same principle, the interaction of a magnetic and the current in a coil of wire.

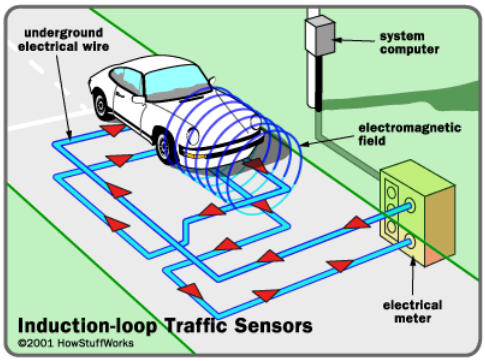



The Speaker – In a speaker, the current through a movable coil of wire, is connected to the speaker cone and a magnetic placed inside this coil. When an electric signal is passed through this wire coil it creates a force due to the current in the wire that interacts with a fix magnet in the speaker, this creates a force that causes the coil to move back and forth with the electric signal being sent to the coil, this back and forth creates the sound wave we hear.

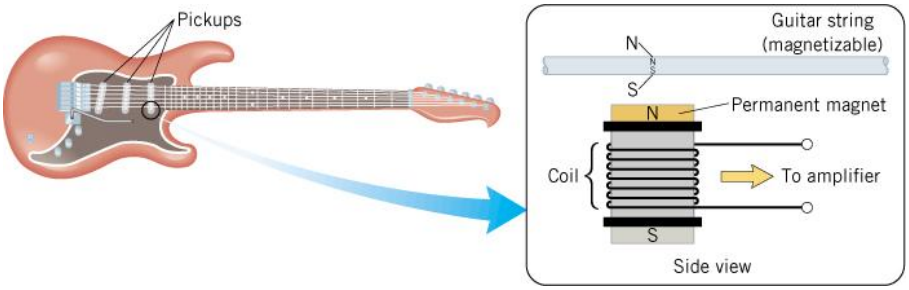
The microphone - has the same set up as the speaker but now the sound that is coming into the microphone (speaker cone) moves the wire coil back and forth around the fixed magnet, this changing magnetic field inside the coil induces a current in the coil at the same frequency as the sound wave incident on the microphone.




**Car traffic light sensor** – When a car (big hunk of iron) drives over a loop of wire in the road, the magnetic field inside the loop of wire is changed (increased) thereby inducing a current in the wire that is sensed at the controls, this induced current/voltage tell the traffic light controller the car is at the light.




**Electric Guitar pickups** – The electric guitar sting is a ferromagnetic material which when plucked vibrates back and forth over a solenoid/coil of wire with an iron core (the pickup), the vibrating string changes the magnetic field inside the coil which induces a current in the coil, this creates an electric signal at the same frequency as the string is vibrating. This electric signal is sent to the amplifier and sound at the vibration of the string is produced.



**Radio Frequency Identification Device (RFID)** – are used to identify products along with any other information stored on an electronic data chip imbedded on the RFID. The chip is connected to a coil of wire (circular pattern around square in image) when this coil is exposed to a changing magnetic field a current is induced and the chip is powered up. When the chip has power, it sends out a signal with any information that is stored on the chip. They can be used in many different applications such as books, boxes, used as ID in road races, as medical tracking devices, etc …



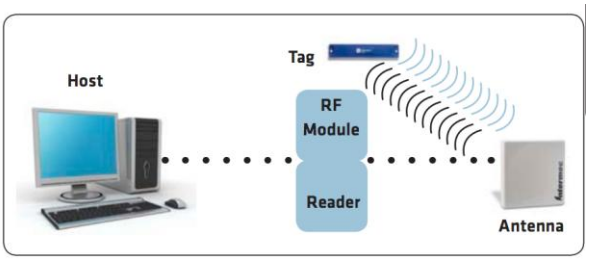
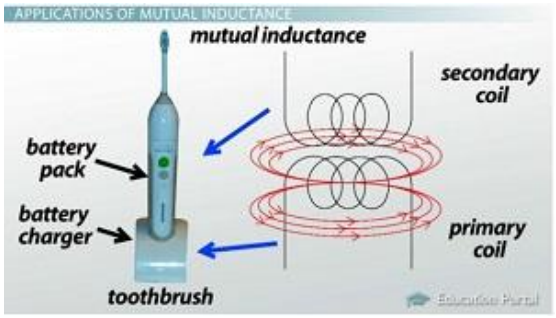


Figure 1: RFID System Components

**Induction charger** – Rechargeable batteries can be charged by mutual induction, a coil of wire is hooked up in series with the battery, when the coil experiences a changing magnetic field current is produced in that coil that produced the energy to charge the battery.



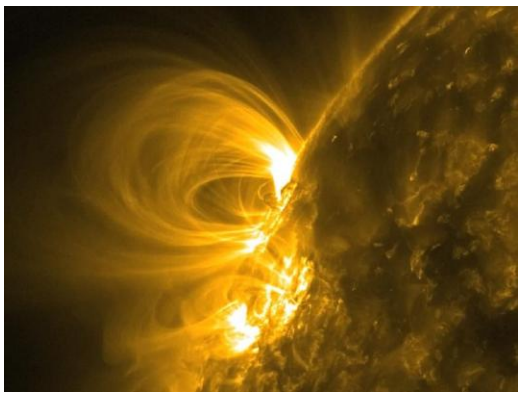

**Solar Flares and the magnetic fields** - A solar flare is an intense burst of radiation coming from the release of magnetic energy associated with sunspots. They are seen as bright areas on the sun, and they can last from minutes to hours. As the magnetic energy is being released from the sun, particles, including electrons, protons, and heavy nuclei, are heated and accelerated in the solar atmosphere. This energy is ten million times greater than the energy released from a volcanic explosion.

> **Sunspots** are areas that appear dark on the surface of the Sun. They appear dark because they are cooler than other parts of the Sun's surface due to strong magnetic fields restricting the movement of the material in the spot.
>
> **Solar flares** are a sudden explosion of energy caused by tangling, crossing or reorganizing of magnetic field lines near sunspots.

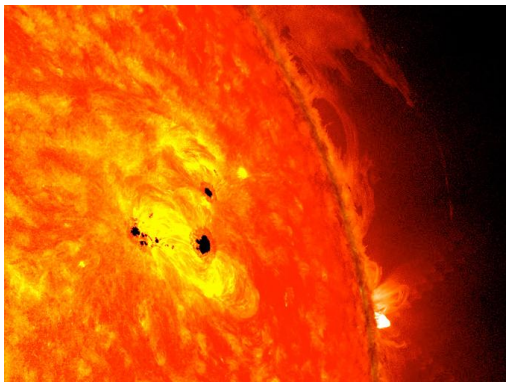
Solar flares extend out to the layer of the Sun called the corona, the corona is the outermost atmosphere of the Sun. The solar flares are loop-shaped because the particles are following the magnetic field lines created by the corona of the sun. These bright loops are located within and

connect areas of strong magnetic field called active regions. Sunspots are located within these active regions, and this is where solar flares occur in this active region. The sun and solar flares launch large amounts of radiation and particles towards the earth call the Solar Wind.

Solar flares on the Sun's surface



Sunspots (dark) and solar flares



The Solar wind is streams of gas, particles and magnetics fields that are ejected from the sun and interacts with the magnetic field of the earth.

The solar wind is composed primarily of charged particles that stream out from the Sun's upper atmosphere, or corona. The main types of particles in the solar wind are:

Protons and electrons – These are the most abundant particles, making up about 95% of the solar wind. These particles have to much energy to bond therefore they impact the earth mostly as separate particles.

Alpha particles ($He^{2+}$ ions) – These are helium nuclei (two protons and two neutrons) and make up about 4–5% of the solar wind.

Trace heavier ions – Such as carbon, oxygen, nitrogen, neon, magnesium, silicon, sulfur, and iron, often in highly ionized states.

These particles travel at speeds ranging from 300,000 m/s (670,000 mph) to over 800,000 m/s (1.8 million mph), depending on whether the solar wind is "slow" or "fast." The solar wind also carries the interplanetary magnetic field (IMF), which is an extension of the Sun's magnetic field into space.

**The strong magnetic fields created by solar flares can cause problems on Earth.**

- Solar flares can disrupt communication from satellites, radio towers and cell towers by inducing currents in the electronics in the systems.
- Solar flares can induce stray voltages in the power grid, due to the large changing magnetic field, if these voltages are large enough, they can blow out transformers and cripple relay stations.